MSML640: Computer Vision
Final Project Report

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1 Introduction

Image classification is a fundamental task in computer vision, with applications ranging from healthcare and autonomous systems to food recognition and dietary analysis. However, training deep learning models for such tasks typically requires large-scale labeled datasets, which are often expensive and time-consuming to collect.

In this project, we explore the use of **transfer learning** to perform food image classification on a relatively small, self-collected dataset. We leverage a pretrained convolutional neural network, EfficientNet-B0 and fine-tune it to classify images into six food categories: *Pizza, Sushi, Burger, Pasta, Salad and Spaghetti.*

A key focus of this work is to analyze how different data enhancement strategies affect model performance. Specifically, we compare four configurations: a baseline model trained on original data, a model trained with data augmentation, a model trained with synthetically generated data using MixUp and CutMix techniques and a combined approach using both synthetic data and augmentation.

In addition to evaluating classification accuracy, we conduct a detailed analysis of model behavior through confusion matrices, error analysis and robustness testing under various perturbations such as noise, blur, occlusion and lighting changes. This allows us to better understand the strengths and limitations of the model in real-world scenarios.

2 Dataset

Dataset Overview

The dataset was divided into three splits: *training*, *validation* and *test*. There was no overlap between these splits, ensuring that the evaluation reflects the model's ability to generalize to unseen data.

The dataset consists of six food categories:

- Burger
- Pasta
- Pizza
- Salad
- Spaghetti
- Sushi

The dataset is balanced, with an equal number of images per class.

Overall Dataset Split

Split	Number of Images
Train	210
Validation	60
Test	60
Total	**330**

Per-Class Distribution

Class	Train	Validation	Test	Total
Burger	35	10	10	55
Pasta	35	10	10	55
Pizza	35	10	10	55
Salad	35	10	10	55
Spaghetti	35	10	10	55
Sushi	35	10	10	55

Since each class contains the same number of images, there is no class imbalance in the dataset. This allows the model to learn all categories uniformly without bias toward any particular class.

Synthetic Data Setup

For experiments involving synthetic data, an additional training set named *train_synthetic* was created. This set includes the original training images along with synthetic samples generated using same-class MixUp and CutMix techniques.

The validation and test sets were kept unchanged across all experiments to ensure a fair comparison between different training configurations.

3 Model Selection

For this project, we selected EfficientNet-B0 pretrained on ImageNet as the backbone model. **EfficientNet-B0** was a good fit for our task because it provides a strong balance between accuracy, parameter efficiency and inference speed.

Since our dataset is relatively small, with six food classes and around 330 total images, training a deep convolutional neural network from scratch would likely lead to overfitting. Using an ImageNet-pretrained model allowed us to transfer useful visual features such as edges, textures, colors and object shapes to the food classification task. These pretrained features are especially helpful for food images because the classes often differ by visual texture, color patterns, ingredients and object structure.

EfficientNet-B0 is also lightweight compared with larger CNN models such as ResNet-50. It has fewer parameters, which reduces training time and lowers the risk of overfitting on a small dataset. At the same time, it still provides strong feature extraction capability because EfficientNet uses compound scaling to balance network depth, width and input resolution efficiently.

Another reason for choosing EfficientNet-B0 was inference speed. Since our project includes a Streamlit demo application, we needed a model that could generate predictions quickly when a user uploads an image. EfficientNet-B0 is small enough for fast inference while still being accurate enough for the classification task.

Overall, EfficientNet-B0 was selected because it is pretrained, efficient, suitable for a small dataset, less computationally expensive than larger models and fast enough for deployment in the demo application.

4 Training Configurations

We fine-tuned the same **EfficientNet-B0** model under four different training configurations to study the impact of data augmentation and synthetic data generation.

Configuration 1: Baseline

The baseline model was trained using only the original training images. All images were resized to 224×224 and normalized using ImageNet statistics. No augmentation or synthetic data was applied in this setting. This configuration serves as a reference to evaluate improvements from other methods.

Configuration 2: Original Data + Augmentation

In this configuration, the original training images were used along with online data augmentation applied during training. The transformations included *horizontal flipping*, *rotation* and *color jitter*.

These augmentations introduce variability in the training data, helping the model learn more robust and invariant features. As a result, the model becomes less sensitive to small changes in image orientation, lighting and color.

Configuration 3: Original Data + Synthesized Data

For this configuration, the training dataset was expanded using synthetic data generation. A new training folder, *train_synthetic*, was created by combining original images with additional images generated offline using same-class MixUp and CutMix techniques.

In *MixUp-style synthesis*, two images from the same class are blended together to create a new sample. In *CutMix-style synthesis*, a patch from one image is inserted into another image of the same class. Since both source images belong to the same category, the generated image retains the correct label.

Configuration 4: Original Data + Synthesized Data + Augmentation

The final configuration combines both synthetic data and online augmentation. The model was trained on the expanded synthetic dataset while also applying augmentation during training.

This approach provides two key advantages: a larger dataset through synthetic generation and additional variability through augmentation. Together, these help improve the model's ability to generalize to unseen data.

5 Results and Analysis

All models were evaluated on the same held-out test set. The test accuracy results are summarized below.

Configuration	**Test Accuracy**
Baseline	86.67%
Original Data + Augmentation	93.33%
Original Data + Synthesized Data	93.33%
Original Data + Synthesized Data + Augmentation	**96.67%**

The best-performing model was **Original Data + Synthesized Data + Augmentation**, achieving a test accuracy of **96.67%**.

The results show that both augmentation and synthesized data improved performance compared to the baseline. Augmentation alone increased the test accuracy from **86.67%** to **93.33%**. Similarly, synthesized data alone also achieved **93.33%**, indicating that the additional generated samples improved generalization.

The highest performance was obtained when both strategies were combined, suggesting that offline synthetic data generation and online augmentation provide complementary benefits for improving model performance.

5.1 Baseline Model

Loss and Accuracy Curves

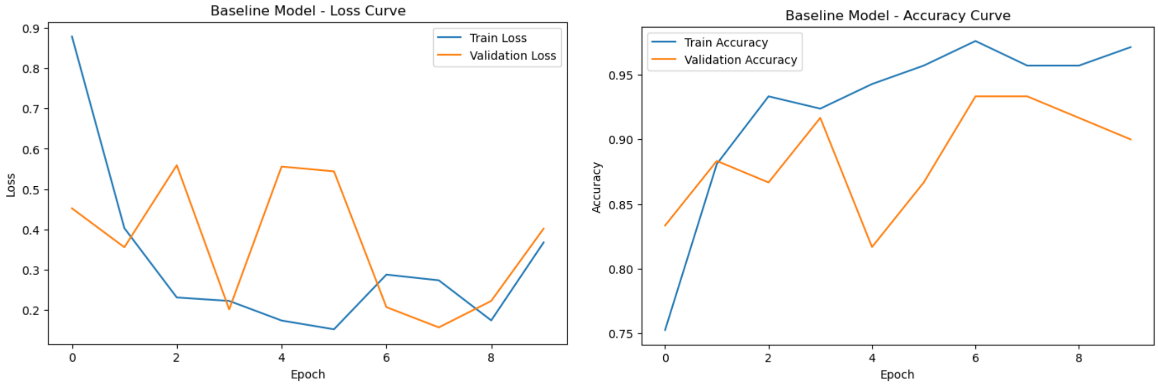

Figure 1: **Baseline Model:** Training and validation loss (left) and accuracy (right) curves across epochs..

The training loss decreases steadily, while the validation loss shows noticeable fluctuations, indicating instability during validation. The training accuracy increases consistently, but the validation accuracy varies across epochs, suggesting mild overfitting and limited generalization.

Confusion Matrix

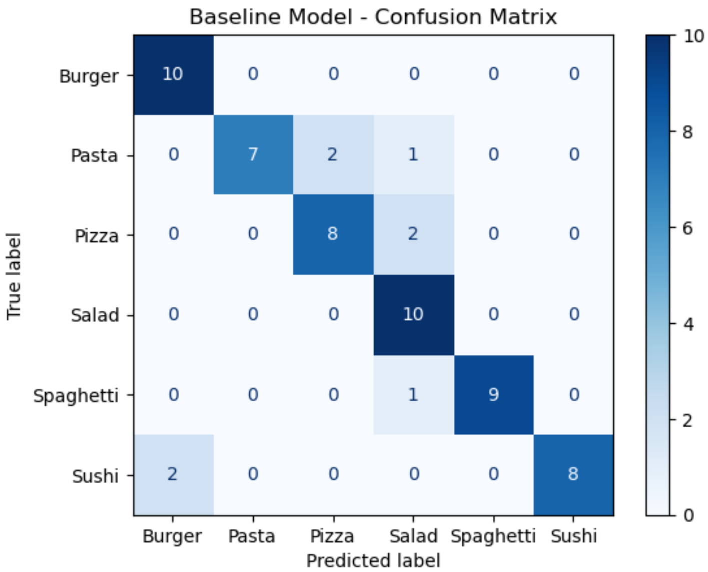

Figure 2: **Baseline Model:** Confusion matrix showing classification performance across all classes.

The model performs well on distinct classes such as Burger and Salad, achieving perfect classification. However, it struggles with visually similar classes, particularly Pasta and Pizza and misclassifies some Sushi images as Burger, indicating confusion between similar textures and appearances.

5.2 Original Data + Augmentation

Loss and Accuracy Curves

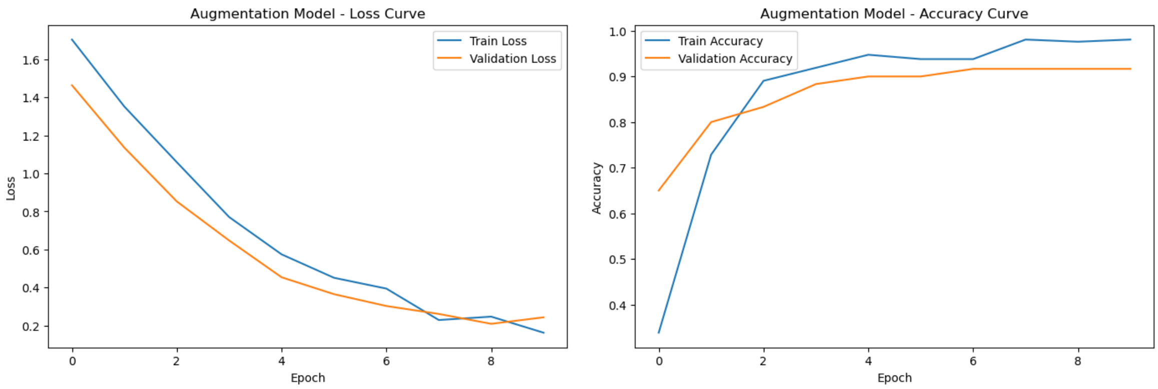

Figure 3: **Augmentation Model:** Training and validation loss (left) and accuracy (right) curves across epochs.

Both training and validation loss decrease smoothly over epochs, indicating stable and consistent learning. Compared to the baseline, the gap between training and validation accuracy is smaller, suggesting improved generalization and reduced overfitting due to augmentation.

Confusion Matrix

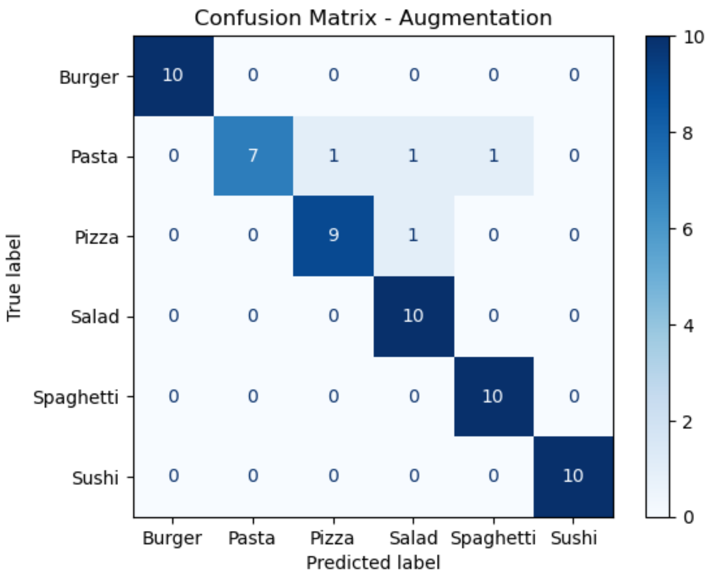

Figure 4: **Augmentation Model:** Confusion matrix showing improved classification performance across all classes.

The model achieves near-perfect classification across most classes, with significant improvement compared to the baseline. Only minor confusion remains within visually similar classes such as Pasta, while classes like Sushi and Spaghetti are classified correctly in all cases.

5.3 Original Data + Synthesized Data

Loss and Accuracy Curves

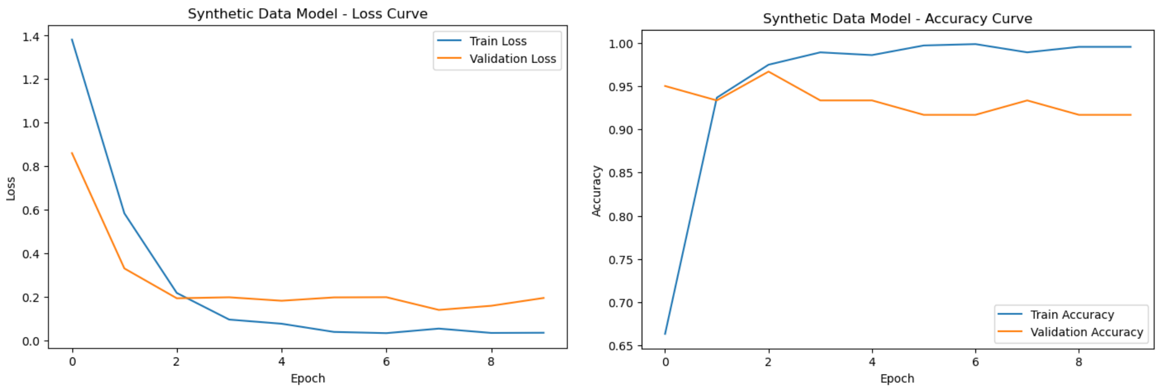

Figure 5: **Synthetic Data Model:** Training and validation loss (left) and accuracy (right) curves across epochs.

The training loss decreases rapidly and reaches very low values, while the validation loss stabilizes after initial improvement, indicating effective learning from the expanded dataset. The training accuracy approaches near-perfect values, while the validation accuracy remains consistently high, suggesting strong learning capacity with slight overfitting.

Confusion Matrix

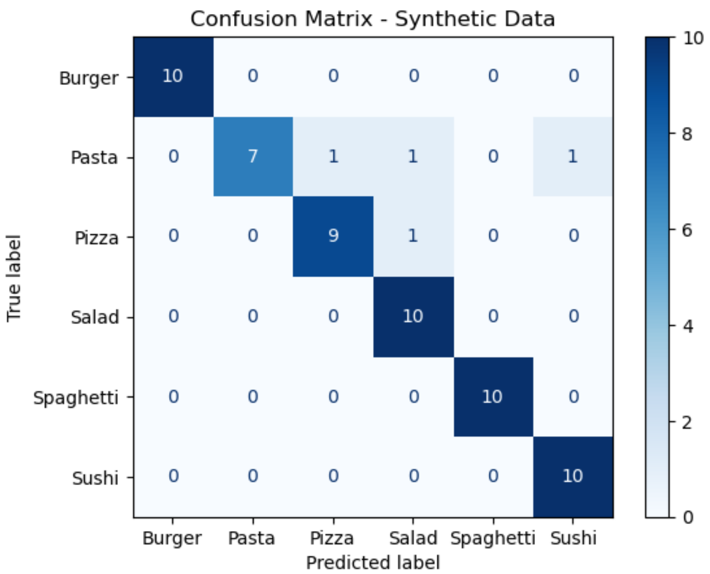

Figure 6: **Synthetic Data Model:** Confusion matrix showing improved classification with minor remaining errors.

The model achieves high accuracy across most classes, with correct classification for Burger, Salad, Spaghetti and Sushi. Minor confusion is still observed for Pasta and Pizza, indicating that visually similar classes remain challenging despite the increased dataset size.

5.4 Original Data + Synthesized Data + Augmentation

Loss and Accuracy Curves

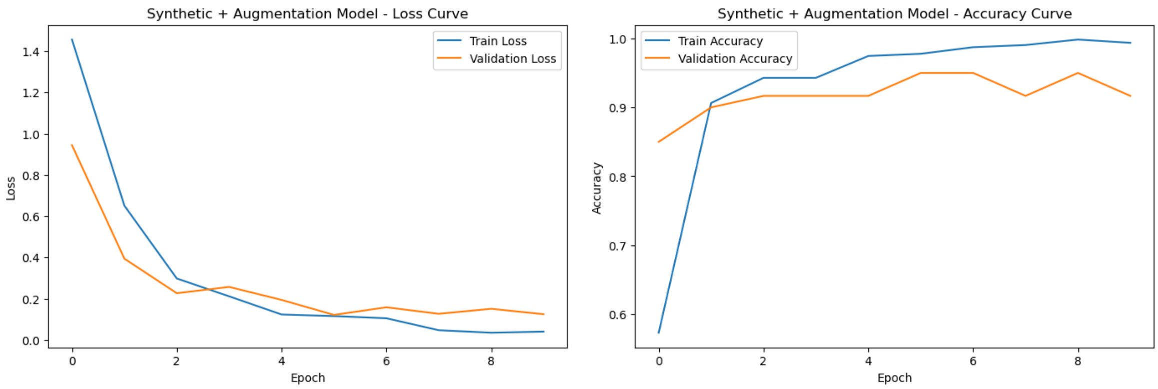

Figure 7: **Synthetic + Augmentation Model:** Training and validation loss (left) and accuracy (right) curves across epochs.

Both training and validation loss decrease consistently and remain stable throughout training, indicating effective and well-balanced learning. The training and validation accuracy curves are closely aligned, showing strong generalization with minimal overfitting compared to previous configurations.

Confusion Matrix

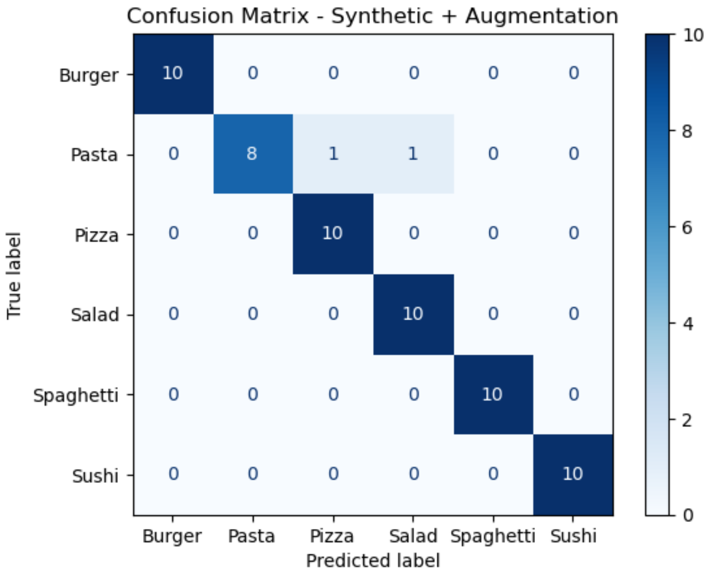

Figure 8: **Synthetic + Augmentation Model:** Confusion matrix showing near-perfect classification performance across all classes.

The model achieves near-perfect classification across all classes, with most categories correctly predicted without error. Only minor confusion remains in the Pasta class, indicating that combining synthetic data and augmentation significantly improves overall class separation and model performance.

5.5 Comparison

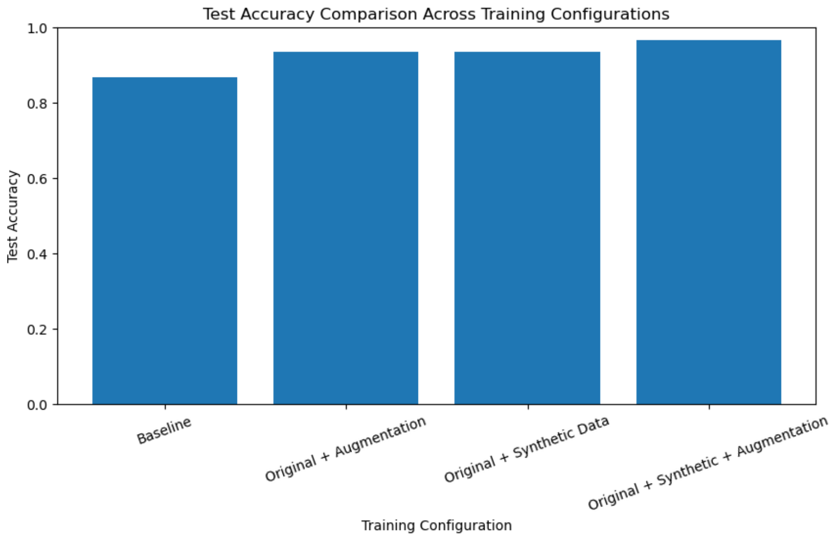

Figure 9: **Test Accuracy Comparison:** Performance of all four training configurations on the test set.

All four configurations were evaluated on the same held-out test set. The baseline model achieved a test accuracy of **86.67%**. Both **Original Data + Augmentation** and **Original Data + Synthesized Data** improved performance to **93.33%**. The best-performing model, **Original Data + Synthesized Data + Augmentation**, achieved a test accuracy of **96.67%**.

The bar graph shows that all enhanced configurations outperform the baseline, indicating that increasing training data diversity improves generalization. Augmentation introduces transformations such as flipping, rotation and color jitter, while synthesized data adds new samples using MixUp and CutMix techniques. Combining both strategies results in the strongest performance.

Confusion Matrix Analysis for Baseline & Best Model

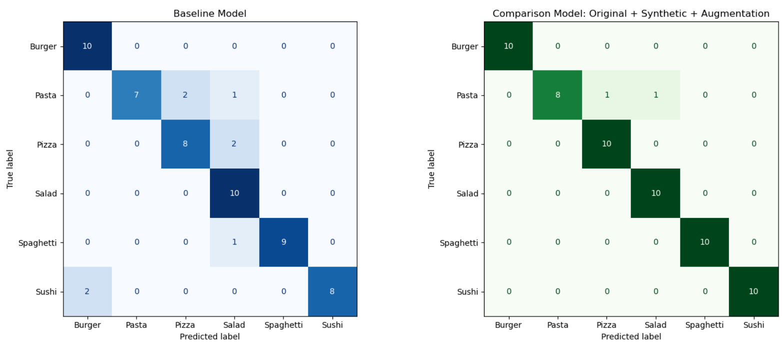

Figure 10: **Confusion Matrix Comparison:** Baseline model (left) vs best model (right).

The baseline model shows higher misclassification rates, particularly between visually similar classes such as Pasta and Pizza as well as some confusion in Sushi classification. These errors indicate difficulty in distinguishing classes with similar textures, colors and ingredients.

In contrast, the best-performing model significantly reduces these errors and demonstrates clearer class separation across all categories. Most classes are classified correctly, with only minor confusion remaining in the Pasta class.

Overall, the accuracy comparison highlights the improvement across configurations, while the confusion matrices illustrate how the best model reduces class-level errors and improves overall classification performance.

6 Error Analysis

The error analysis focuses on identifying which classes were misclassified, the types of errors made by the model and the underlying reasons for these mistakes.

Most errors occur between visually similar classes. For example, *Pasta* is one of the most challenging classes because it shares visual characteristics with Pizza, Salad and Spaghetti. Pasta images may include sauce, vegetables, cheese and mixed textures that resemble other food categories. Similarly, Pizza can sometimes be confused with Pasta or Salad when toppings and color distributions dominate the image.

Baseline Misclassified Examples

Index	True Class	Predicted Class	Confidence
10	Pasta	Pizza	0.9911
16	Pasta	Pizza	0.6647
17	Pasta	Salad	0.9987
27	Pizza	Salad	0.8375
28	Pizza	Salad	0.6324
46	Spaghetti	Salad	0.9254
56	Sushi	Burger	0.9603
59	Sushi	Burger	0.6672

Best Model Misclassified Examples

Index	True Class	Predicted Class	Confidence
16	Pasta	Pizza	0.8510
17	Pasta	Salad	0.9734

Error Summary by Class

Class	Baseline Errors	Best Model Errors
Pasta	3	2
Pizza	2	0
Spaghetti	1	0
Sushi	2	0

The most likely reasons for these errors include:

- Similar color distributions across classes
- Shared ingredients such as vegetables, cheese and sauces
- Loss of fine details due to resizing images to 224×224
- Limited dataset size
- Fine-grained differences between visually similar categories

The baseline model produces more errors due to limited exposure to diverse training data. In contrast, the best-performing model significantly reduces these errors because the combination of synthesized data and augmentation introduces a wider range of variations.

However, some classes such as Pasta remain challenging due to subtle visual differences. This highlights that even with high overall accuracy, error analysis is essential to understand the limitations of the model and identify areas for further improvement.

7 Robustness Analysis

Robustness testing was performed on the best-performing model (*Original Data + Synthesized Data + Augmentation*) to evaluate how well it handles distribution shifts and image perturbations. The model was tested on both clean images and perturbed versions of the test images.

Perturbation Types

Perturbation	Description
Clean	Original test images
Gaussian Blur	Blur applied to simulate out-of-focus images
Occlusion	A black box added to hide part of the image
Gaussian Noise	Random noise added to simulate image corruption
Lighting Change	Brightness modified to simulate lighting variation

Robustness Results

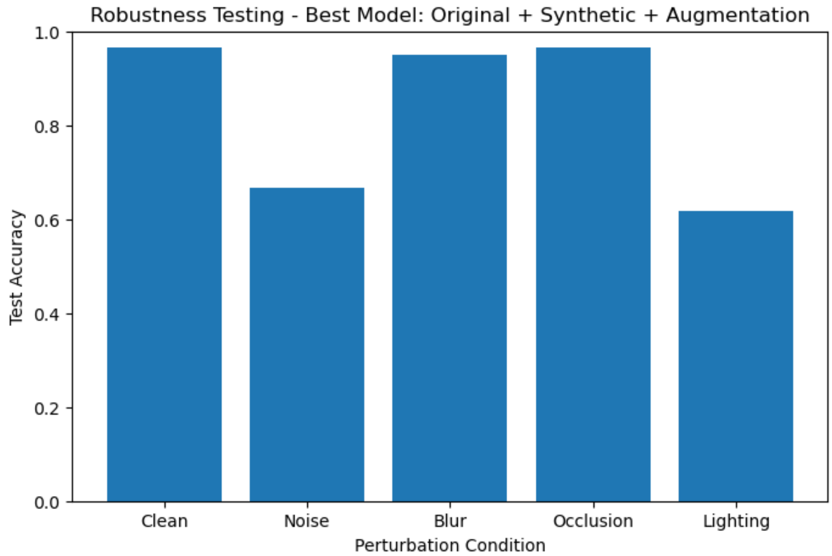

Figure: Accuracy under different perturbation conditions

The model demonstrates strong robustness to *blur* and *occlusion*, with only minor drops in accuracy. This indicates that the model has learned meaningful global features and does not rely solely on specific local regions of the image.

However, performance drops significantly under *noise* and *lighting changes*. This suggests that the model is sensitive to pixel-level distortions and variations in color distribution. In particular, the drop in performance under lighting changes is important, as real-world food images often vary in brightness, shadows and camera conditions.

These results highlight that while the model generalizes well under structural distortions, it is less robust to changes in image quality and illumination.
Future improvements could include stronger lighting-based augmentation, noise injection during training and collecting more diverse real-world images captured under varying environmental conditions.

8 What Worked, What Failed, and Why

8.1 What Worked

Transfer learning was highly effective for this project. The EfficientNet-B0 model, pretrained on ImageNet, adapted well to the food classification task despite the relatively small dataset. The pretrained features enabled the model to learn meaningful visual patterns quickly.

Data augmentation also improved performance significantly. The augmentation configuration increased test accuracy from **86.67%** to **93.33%**. This demonstrates that applying transformations such as rotation, horizontal flipping and color jitter helped the model generalize better to unseen data.

Synthesized data further contributed to performance improvements. The configuration using synthetic data alone also achieved **93.33%** test accuracy, indicating that generating additional training samples introduced useful variation into the dataset.

The best performance was achieved by combining both synthesized data and augmentation. This configuration reached **96.67%** test accuracy. The improvement suggests that the two strategies are complementary: synthetic data increases dataset size, while augmentation introduces dynamic variability during training.

8.2 What Failed

The baseline model performed reasonably well but showed weaker generalization compared to the augmented and synthetic-data models. Its lower test accuracy indicates that training only on the original dataset limited exposure to diverse visual variations.

Although synthetic data improved performance, it did not outperform augmentation when used alone. This may be because synthetic images were generated from existing samples, which increases variation but does not fully capture real-world diversity.

The model also struggled under certain perturbations during robustness testing, particularly under *noise* and *lighting changes*. These conditions caused a noticeable drop in accuracy, indicating sensitivity to pixel-level distortions and color variations.

8.3 Why These Results Occurred

The baseline model benefited from pretrained ImageNet features, but the limited dataset size restricted its ability to generalize. Data augmentation improved performance by exposing the model to multiple transformed versions of the same images during training.

Synthetic data increased the effective size of the training dataset by generating additional samples, which helped improve learning. However, since these samples were derived from existing images, their diversity was still somewhat limited.

The combined model performed best because it leveraged both strategies simultaneously: a larger training set from synthetic data and additional variability from augmentation. This combination allowed the model to learn more robust and generalized representations.

The weaknesses observed in robustness testing are primarily due to the nature of the training data. Since the model was trained mostly on clean images, it was not sufficiently exposed to noise or extreme lighting variations. As a result, it struggled when these conditions were introduced during testing.

9 Conclusion

This project explored the use of transfer learning for food image classification by fine-tuning an ImageNet-pretrained EfficientNet-B0 model on a six-class dataset. Four training configurations were evaluated to understand the impact of data augmentation and synthesized data on model performance.

The baseline model achieved a test accuracy of **86.67%**. Both augmentation and synthesized data individually improved performance to **93.33%**. The best-performing configuration, which combined original data, synthesized data, and augmentation, achieved **96.67%** test accuracy.

The results demonstrate that both augmentation and synthesized data contribute to improved generalization, with the combination of the two providing the most benefit. Confusion matrix analysis and error analysis revealed that most remaining errors occurred between visually similar food categories. Robustness testing showed that the model performed well under blur and occlusion but remained sensitive to noise and lighting variations.

To demonstrate practical usability, a Streamlit-based application was developed. The application allows users to upload food images and view predicted classes along with confidence scores, showcasing the effectiveness of the trained model in a real-world setting.

All graphs, plots, confusion matrices and robustness visualizations used in this report are supported by corresponding source code committed to the GitHub repository.